Zenith: Dated Evaluation and Controlled Updates of a Reasoning Graph

Implementation and retrospective evaluation

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Abstract

Zenith evaluates changes to a reasoning graph using dated inputs, native output contracts and complete paired forecast support. Its executor binds runs to source, graph, policy and provider identities; diagnostics distinguish recorded failures from hypotheses about their cause. A proposal compiler can change one node-policy field for controlled replay. A historical price-only study compared one manually frozen persistence/scale rule with a model baseline over 20 August and five later October 2015 dates. On the later dates, candidate mean relative price error was 0.8875%, versus 0.9966% for the model baseline and 1.0453% for no change. On development dates, no change outperformed both model policies. The 100-call study demonstrates pipeline feasibility and a descriptive intervention comparison. Previously examined history, a modern model and limited support prevent claims of prospective accuracy, learned improvement or novelty. The learning question is whether measured development feedback produces a forecast-node skill that improves later use of financial evidence. An offline A-share compiler now admits 136 five-session issue batches across ten stocks for a comparison of information, outcome-trained skills, hindsight-trained skills and reflected memory. This later study has not executed model inference.

1 Question and scope

The intended system answers three questions. What should be forecast from information available at issue time? What happened at the declared horizon? What does the recorded attempt suggest changing? The first is a forward reasoning task, the second a measurement task, and the third a diagnosis and learning task. Price error alone cannot answer the third: a miss can reflect unavailable information, restricted context, an unsupported inference, response variation or an unpredictable event.

Zenith is a standalone extraction of an existing Lab–Oracle–Refinement system. The original Lab combines stock and ETF prices, macro observations, news and other feed lanes. Reasoning nodes interpret the supplied evidence through instructions and typed output contracts; they can consider competing explanations rather than execute a fixed

*AI agents researched and drafted this manuscript under Will Cook's direction. The source and experimental receipts define the evidence reported here.

indicator rule. Lab emits a thesis checkpoint (CP1) and forecast checkpoint (CP2); Oracle resolves later observations and critiques the frozen attempt. A retained dossier runner and a newer reflection/cohort route implement different revision paths.

Graph-based reasoning is established prior art [2]. The local question is whether measured development feedback can improve a persistent reasoning skill on later evidence under controlled replay. The completed study uses a three-node price-only replacement, with one manually frozen skill rule. It establishes execution and comparison boundaries; it does not test the full multi-source graph or answer the learning question.

2 Implemented mechanism and its guarantees

2.1 Native execution and evidence boundaries

A node declares dependencies, instructions and an output schema. A reasoning policy can add a short skill rule, set nonnegative incoming-edge weights or limit context characters. Base instructions and output contracts remain binding. Canonical artifacts are retained; context views contain full artifacts, declared excerpts or omission markers. Receipts record source hashes, retained characters and missing dependencies.

Declared edges are scheduled inputs. Effective requests also contain injected contracts, temporal substitutions and configured context. Agent-provider handoff packets retain complete prompts; the paid adapter records prompt hashes, and engine attempt sidecars retain response text with at most 4,000 prompt characters. A graph path therefore directs inspection without proving actual model consumption or causal responsibility. Native output validation applies to both paid responses and imported agent responses. Paid cohorts require the same requested provider endpoint in both arms and check returned provider identity against its catalog binding.

Observation time, first availability and acquisition time are distinct. Forward evidence must belong to the subject-side class and have declared availability no later than the issue. Unknown availability remains uncertain. Conservative historical release rules do not authenticate publication clocks or price vintages. Later truth is separated from forward execution.

The conditional Lean model proves composition when the constituent skills satisfy their assumed relations, and expresses availability and rank-based acyclicity constraints. The Python checker verifies declared ports, evidence classes, source hashes and finite ranks. Neither establishes that a natural-language skill implements its relation, that a timestamp is authentic, or that the forecast is accurate.

2.2 Measurement and proposals

Each forecast names its target, horizon, issue and target prices, basis and currency. Before maturity it is pending; missing or incompatible truth is a data gap. For issue i, fixed target set S_i, predictions q_{is} and positive realized prices y_{is}, the cohort loss is

$$L_i = \frac{1}{|S_i|} \sum_{s \in S_i} \frac{|q_{is} - y_{is}|}{y_{is}}, \qquad L = \frac{1}{n} \sum_{i=1}^{n} L_i.$$

Targets are averaged within issues, then issues receive equal weight. Complete comparison requires every included arm to have the exact frozen issue/target support. Partial runs expose coverage without aggregate, per-issue or paired scores. Complete receipts retain $L_i^{\text{candidate}} - L_i^{\text{baseline}}$; negative values favor the candidate.

This is a point-price loss, not a trading return. No-change and past-one-session momentum use the same support and loss. For positive outcomes with finite inverse moment, absolute percentage error elicits an inverse-price-weighted median, generally distinct from a conditional mean [6]. Current instructions ask for a target price without naming that functional. Any subsequent change to the elicited quantity or loss must be declared before execution, rather than rescoring the existing study opportunistically.

Diagnostics bind artifact presence, validator rejection and context-delivery conditions to recorded bytes and receipts. Missing or unbound evidence stays unknown; observer failure cannot change cohort acceptance. A large final error can select a trace for inspection but cannot certify missing data, data-access failure or faulty reasoning as its cause. A hindsight reviewer may inspect realized outcomes and later news to form a hypothesis. Its record must separate checkable defects, relevant information available before issue but omitted, post-issue events and unresolved explanations. Published explanations or reviewer labels supply no causal ground truth. Only admissible pre-issue evidence may enter a repaired forward run; its benefit needs a new measured comparison.

The reflection route binds development membership, feedback and selected node artifacts plus available agent-provider exchanges. It does not consume all paid debug sidecars; without a handoff packet the selected input prompt is unavailable. An agent supplies a hypothesis and intervention. The compiler checks the measured baseline, changes one skill rule, context budget or edge weight through the policy validator, and records before/after identities. Compilation neither executes nor accepts the candidate. An input intervention needs a separately declared acquisition/ablation protocol. Coupled changes can be tested through baseline/A/B/A+B replay; single-field changes have no general optimality guarantee.

3 Completed historical experiment

3.1 Fixed protocol and provenance

The experiment uses StockNet price evidence [12], with AAPL, MSFT and GOOG fixed throughout. It supplies 20 past exchange sessions of OHLCV and predicts the next exchange-session close. Development contains the first 20 common mature August 2015 issue dates; later assessment contains the first five October dates. The native cohort freezes candidates, inputs and source identities, separates split membership by calendar week, checks the declared one-day embargo, and seals final assessment before opening its truth.

Both arms use the same three-node graph, model, provider and 24,000-character context caps. Their sole difference is one CP2 skill rule: use current close as a persistence prior, separate direction from move size, and shrink movement toward zero when past-return scale evidence is weak, while preserving target and grounding re-

quirements. This rule was manually fixed before development scoring, not generated by measured feedback. The baseline and candidate policy identities are frozen in the cohort at prepared source revision `01522cc`, hash prefix `bd5224387591`; full identities are in `paper/study-evidence.json`.

The August outcomes had been exposed in an earlier flat-input study, and the October dates had been examined in deterministic benchmark work. This freeze supplies a controlled historical comparison, not a pristine holdout. The modern model may also encode historical outcomes. Valid input timestamps cannot remove pretrained look-ahead knowledge; anonymization controls can test some effects without proving absence of memorization [5].

Four compatibility pilots used 1 July 2015, excluded from the study. Nova and Qwen responses failed length or reference validation; an unpinned DeepSeek request returned HTTP 429 without usage or known charge. An audited recovery preserved that failed attempt and reservation. A fresh DeepSeek request through `deepinfra/fp8`, reported as DeepInfra, completed both reasoning nodes. These six attempts yielded five responses and USD 0.004173065 in known charges. Acceptance measures contract compatibility, not forecast accuracy. `paper/calibration-evidence.json` and `paper/README.md` retain all pilot outcomes, tokens, costs and implementation-hash matches; manifests record no Git HEAD, so matches do not establish a unique executed revision.

3.2 Complete common-support results

All study requests named `deepseek/deepseek-v4.1-flash`, requested endpoint `deepinfra/fp8`, and returned the matching model through reported DeepInfra. Development completed 40 episodes across 20 dates and four calendar weeks; later assessment completed 10 episodes across five dates and two weeks. Each arm has three scored targets per date. No study calls failed or had unknown cost. The later receipt matches the state sealed once under `native-deepinfra-final-v1`.

Table 1: Complete study loss as percentages, $100L$. Each issue has three fixed targets; deterministic references share support but are not model arms.

Phase	Dates	Baseline	Candidate	No change	Momentum
Development	20	2.1000	1.9142	1.8254	2.3756
Later assessment	5	0.9966	0.8875	1.0453	0.9038

Development candidate-minus-baseline mean issue loss was −0.00185743, lower on 14 dates and higher on six. The candidate remained worse than no change, but passed the frozen rule for a later candidate proposal. Later mean paired loss was −0.00109107, lower on three dates and higher on two; the candidate was below all three reference forecasts on this support. The native final decision remained `insufficient_issues`: five dates are below its 20-issue proposal gate, and the receipt permits no live mutation. These routing thresholds are not significance tests.

Development used 80 responses at USD 0.0992369112; later assessment used 20 at USD 0.0244918688. The complete study cost USD 0.1237287800. With calibration, the

ledger at study close had 106 attempts, 105 responses and USD 0.1279018450 in known charges, plus the earlier unknown-cost HTTP 429. It retained USD 1.137895220 in reservations and was not halted; reservations are not actual charges. Exact scores, paired losses, target coverage and accounting remain in the immutable study export. This small retrospective comparison has no unchanged-baseline replicates and estimates no uncertainty, profitability or prospective forecasting advantage.

4 From a manual intervention to measured learning

4.1 Closest prior art and originality boundary

Financial hindsight learning has close predecessors. FinMem uses future price direction during training to construct experiences retrieved at later decisions [13]. FinCon revises investment beliefs from completed episodes [14]. Learning Using Privileged Information formalizes a teacher with information unavailable to the deployed predictor; its exchange-rate example explicitly considers future observations in historical training [9]. These precedents motivate a controlled information comparison, but supply no guarantee for language-model skill revision.

QuantAgent's 2024 writer/judge loop retrieves experience, while an outer market-data evaluation retains signal code, scores and feedback [10]. ATLAS's Adaptive-OPRO updates static trading instructions from prompt–performance history and recent outcome summaries while protecting run-time inputs and interfaces [7]. These precedents already connect financial outcomes to persistent program or instruction changes. Their alpha-factor and portfolio evaluations differ from the point-price loss in Table 1; adjacent windows under different prompts do not isolate the causal effect of an edit.

Trace propagates executed subgraphs and output feedback to trainable parameters; OptoPrime proposes prompt, code or numerical updates [4]. GEPA reflects on execution feedback and searches prompt candidates [1]. Zenith presently compiles a reviewed node-policy intervention for native replay. It does not implement either optimizer or unrestricted generated programs. A skill change on a fixed graph is policy adaptation, not learned topology. Dated admission, diagnostics and frozen comparison are the local implementation focus; no new learning algorithm or established novelty is claimed. Reviewed versions, software pins and broader comparisons remain in the source manifests, rather than forming a citation inventory here.

4.2 Compiled A-share experiment, without model results

The next experiment preserves two language-reasoning nodes: interpretation and forecast. The forecaster receives the original evidence as well as the saved interpretation, and only its short skill is mutable. The base skill distinguishes reported growth from surprise relative to a supplied expectation, new facts from repeated reports, economic transmission from causal certainty, and the effect remaining after entry from an overnight response already missed.

The pinned TraderHarness tables provide a fixed ten-stock panel. After a twenty-session history window plus a preceding close within the continuous February 2023 onward interval, the compiler schedules 138 nonoverlapping five-session windows. Two

are purged across phase boundaries. All 136 remaining windows have complete historical bars, outcome bars and CSI 300 endpoints: 41 training issues in 2023, 47 validation issues in 2024 and 48 assessment issues in 2025. These are 1,360 asset allowances, not 1,360 independent market histories. The earlier numerical anchor uses a fixed 2021 fit schedule.

Each asset receives at most four issuer-tagged items and two common headlines from the preceding seven calendar days. In 452 asset windows there is no issuer-tagged item. Announcement titles and derived headlines lack full-report context; an upstream issuer token can also identify a related listed company incorrectly. Financial-statement vintages remain excluded. News epochs are interpreted explicitly in UTC; day-precision announcements become available at the next exchange open. These publication rules do not certify historical wording or freedom from pretrained-model knowledge.

The target is next-session open to fifth-session close stock gross wealth return minus the matching CSI 300 price-index return. Entitled shares and unreinvested pre-tax cash enter the wealth mark; an ex-date-open purchaser receives no earlier-holder distribution. Historical wealth paths use only actions occurring by their own issue cutoff. This mixed-basis difference is not an executable hedged portfolio return.

Information arms compare price, price plus event metadata, and admitted text, with the same issuer-code cues and separate interpretations. Learning arms compare unchanged, outcome-trained, hindsight-trained and feedback-blinded skills, plus bounded retrieval of reflected experiences. Outcome and hindsight teachers receive identical forward traces and per-asset residuals; hindsight alone adds reporting from the completed outcome interval. Lessons cannot become available before either their outcome or the latest report they use. Each review states an observable trigger, conditional lesson, uncertainty and disconfirmation condition. One candidate per skill arm competes with unchanged on validation; unchanged wins ties. The memory comparator is an implementation inspired by the mechanism, not a reproduction of FinMem or FinCon, and its larger context allowance precludes an equal-token efficiency claim.

Primary MSE averages stocks within each issue, then weights issues equally. For a single issue, predicted returns $\hat{y}_s$ and outcomes y_s satisfy

$$\frac{1}{S}\sum_s (\hat{y}_s - y_s)^2 = (\bar{\hat{y}} - \bar{y})^2 + \frac{1}{S}\sum_s \left[(\hat{y}_s - \bar{\hat{y}}) - (y_s - \bar{y})\right]^2.$$

The decomposition separates error in the common level from error in relative stock magnitudes. Ranking is checked separately. Training-fitted calibration and a skill-on-metadata diagnostic test simpler explanations of improvement. Missing forecasts make aggregate error unavailable. Paired issue losses use descriptive block intervals, without claiming multiplicity-adjusted superiority. Seven prepared behavioral probes, motivated by CheckList [8], test supplied expectations, paraphrases, duplication, absent support and an already observed price response; they have no historical counterfactual labels.

The compiler, teacher clocks, stage provenance, selection and accounting have offline contract tests. The admission receipt and exact protocol are `docs/ashare-admission.json` and `docs/ashare-study.md`. The complete plan allows at most 1,213

core model calls plus 14 optional probe calls, exceeding the previous 600-call authorization. Execution remains held. No new information-use, learning, trading or novelty result follows from this compile.

4.3 Controlled routing validation

A separate deterministic benchmark composes the real context-policy owner with a Decimal split-adjusted-price calculation and an independently implemented Fraction oracle, without model calls. Across seven scenarios, a finite selector scores unchanged policy and one-edge restorations witnessed in omission receipts on 20 training cases; unchanged wins ties. The selected policy is fixed before 64 assessment cases with disjoint price/factor ranges are generated. Execution accepts only sources and cutoff, rejects oracle labels, validates policy hashes and preserves canonical inputs. Restoring the relevant split edge yields 64/64 contract-correct numeric answers (absolute tolerance 10^{-25}), versus 0/64 for unchanged or irrelevant-edge restoration. Absent, future and conflicting sources yield 64 correct abstentions and zero numeric answers each; healthy and irrelevant-omission controls retain unchanged policy and 64 correct answers. A faulty arithmetic operator stays wrong in all 64 cases under every routing arm. Restoration increases delivered context, so this establishes no fixed-budget efficiency. Published seeds, constructed faults and a restricted candidate set make this integration/control validation, not an LLM skill-learning result, causal discovery or financial forecasting evidence. The source-bound receipt is `docs/repair-controls.json`.

5 Limits and reproduction

The completed result establishes native execution and a bounded historical intervention comparison. It does not establish automatic credit assignment, causal diagnosis, generalization of a learning procedure, usefulness of the full Lab graph, market efficiency or profitability. Context restrictions, missing evidence and reasoning can interact; an intervention measures conditional sensitivity without uniquely identifying the original cause. A modern model's stored knowledge, previously inspected windows, serially related dates and three selected assets limit broader interpretation. Generation and selection are fitting [3]; a later learning study must freeze proposals before outer assessment and disclose lifetime exposure. An unchanged-rule repeat estimates model variation; a feedback-free proposal control distinguishes feedback value from making another proposal. Neither is present here. Repeated search also needs dependence-aware uncertainty over its full comparison family, for which White's Reality Check is one framework [11], not a statistic implemented or estimated in this study.

The compiled learning study follows source audits that withdrew a sparse AAPL news schedule. Its fixed population, limited titles, uncertain entity links, historical inspection and unknown model pretraining constrain any future interpretation. A failed text contrast would not show that financial language is generally uninformative. Prospective capture is required for stronger claims about unseen evidence; executable trading requires a separate fixed allocation and complete market-specific accounting.

The repository contains executor/schema source and text-free evidence projections, not redistributed market text or private provider packets. Candidate code is trusted

local code; the runner is not a security sandbox. Client-side reservations and reported usage do not guarantee provider billing.

The source-owner map is in `paper/README.md`; `docs/reproducing.md` owns environment and offline checks. Source and finance manifests retain primary versions, software pins and adoption decisions.

From an editable source checkout, `python -m zenith validate` and `python -m zenith plan lab` inspect the authored graph without inference. The paper exporter reads existing receipts without evaluating, selecting or sealing. Its validation accepted the actual complete phase reports and rejected copied missing-support, altered-loss and invalid-selection probes while preserving the original artifacts. Every measured claim must bind source, inputs, graph/policy and model/provider receipts; a code citation alone cannot reproduce the comparison.

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